1 Introduction

This report presents a laboratory experiment conducted as part of the ELE2200 course. The objective of the experiment is to analyze the behavior of the system under study and to compare experimental results with theoretical predictions.

The report is structured as follows. Section 2 presents the theoretical background. Section **??** discusses the experimental results. Finally, Section 3 concludes the report.

2 Theoretical Background

The theoretical analysis is based on standard principles introduced in class. The system is modeled using ideal assumptions in order to obtain a closed-form analytical expression.

Let the input variable be denoted by $x(t)$ and the output by $y(t)$. Under linear operating conditions, the system behavior can be expressed as

$$y(t) = Kx(t),$$

where K is a constant gain parameter.

These theoretical results will be used as a reference to interpret the experimental data obtained in the laboratory.

3 Conclusion

In this report, the theoretical and experimental aspects of the laboratory experiment were presented. The theoretical model provided a useful approximation of the system behavior.

Overall, the experimental results were consistent with the theory within expected measurement uncertainties. Possible sources of error include component tolerances and measurement noise.

Future work could involve refining the model assumptions or extending the analysis to non-ideal operating conditions.